Alzheimer Detection — Phase 2 Final Report

Project folder: C:\Users\Ayush\Alzheimer_Project
Model folder: C:\Users\Ayush\Alzheimer_Project\models
Generated: 2025-10-18 21:59:14

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1. Classification Metrics

	precision	recall	f1-score	support
Mild Dementia	0.543454	0.296281	0.383491	5002.00000
Moderate Dementia	0.969697	0.196721	0.327087	488.00000
Non Demented	0.870688	0.953661	0.910288	67222.00000
Very mild Dementia	0.607232	0.441676	0.511389	13725.00000
accuracy	0.830050	0.830050	0.830050	0.83005
macro avg	0.747768	0.472085	0.533063	86437.00000
weighted avg	0.810477	0.830050	0.813170	86437.00000

2. Confusion Matrix

	0	1	2	3
0	1482	0	2116	1404
1	64	96	177	151
2	746	3	64107	2366
3	435	0	7228	6062

3. Model Layers Summary

Model summary file not found. If you have a Keras model object available, consider saving model.summary() to outputs/model_summary.txt and rerun.

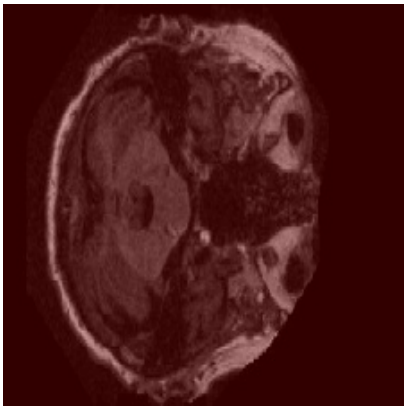
Meanwhile we list detected Grad-CAM class folders below:

- Mild Dementia
- Moderate Dementia
- Non Demented
- Very mild Dementia

4. Grad-CAM Visualizations

Below are representative Grad-CAM images (per class). Captions are file names.

Class: Mild Dementia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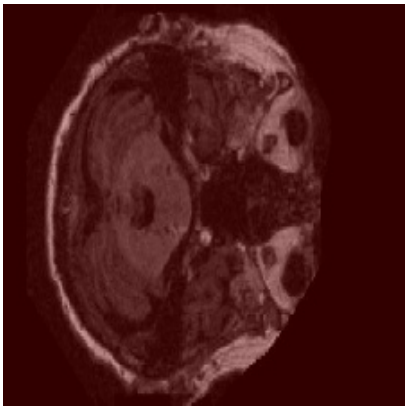
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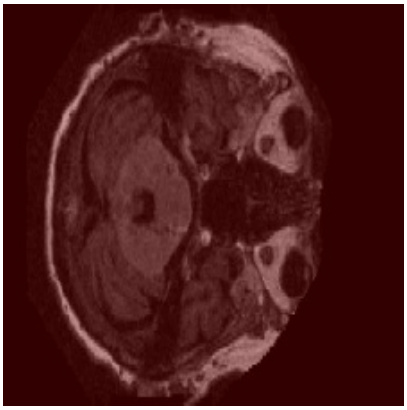
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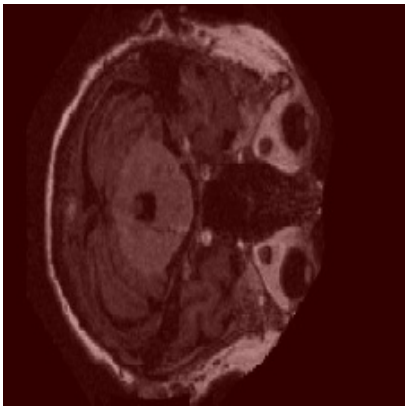
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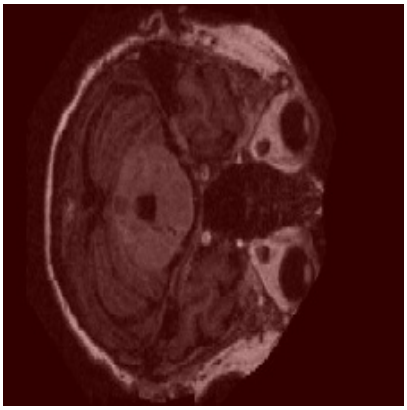
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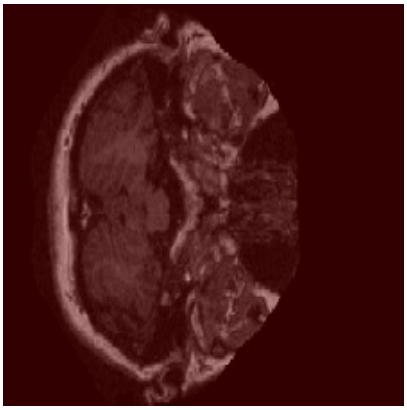
Class: Moderate Dementia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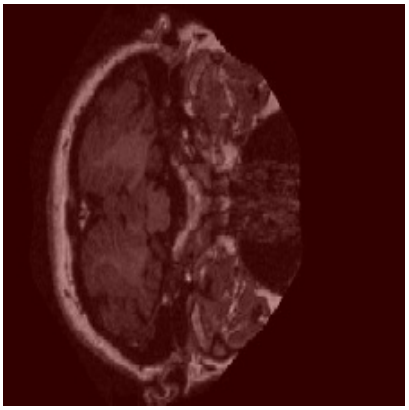
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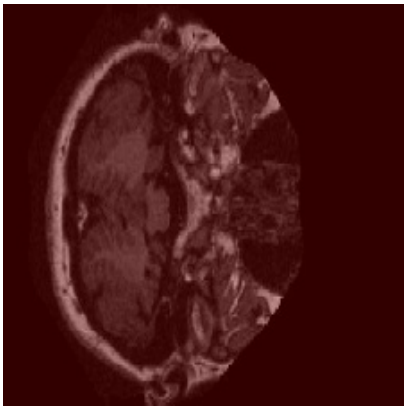
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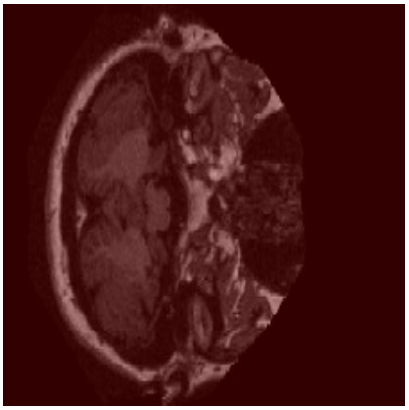
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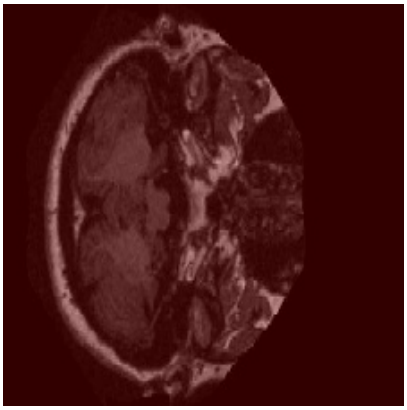
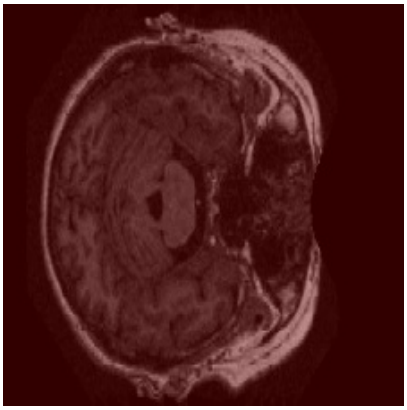

gradcam_OAS1_0308_MR1_mpr-1_104.jpg

Class: Non Demented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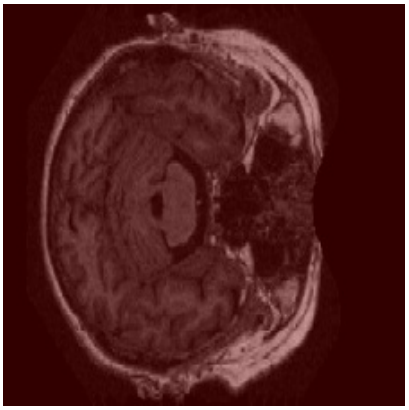
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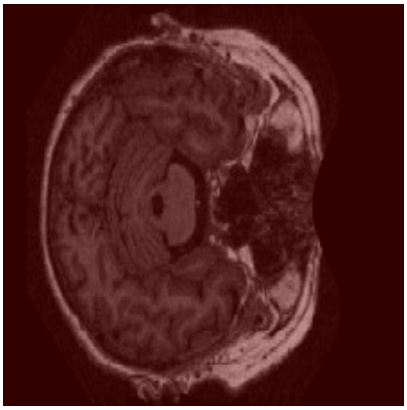
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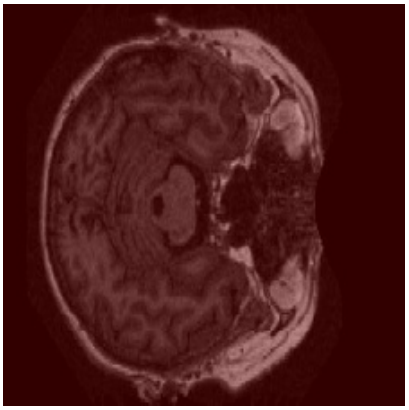
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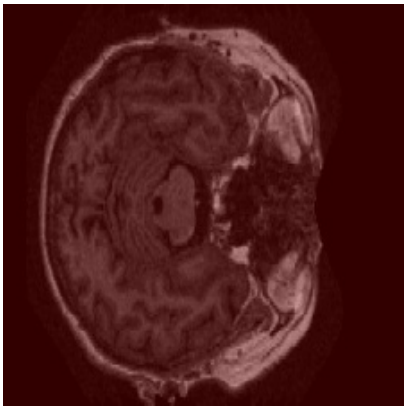
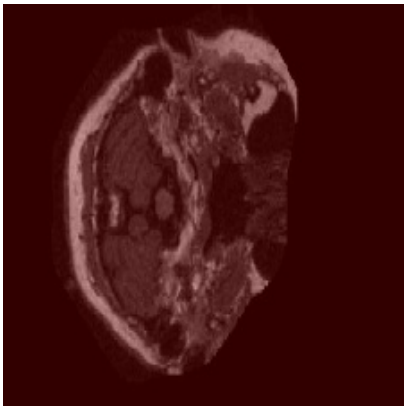

gradcam_OAS1_0001_MR1_mpr-1_104.jpg

Class: Very mild Dementia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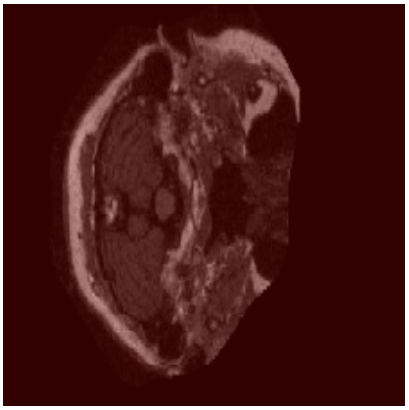
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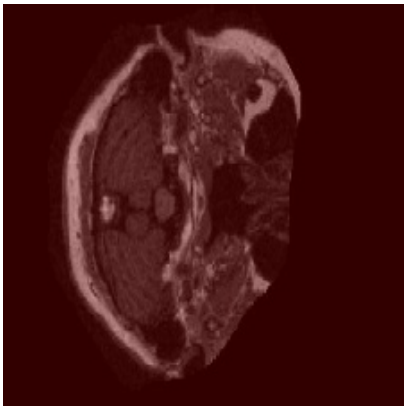
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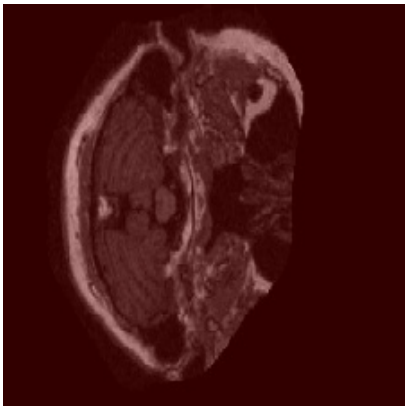
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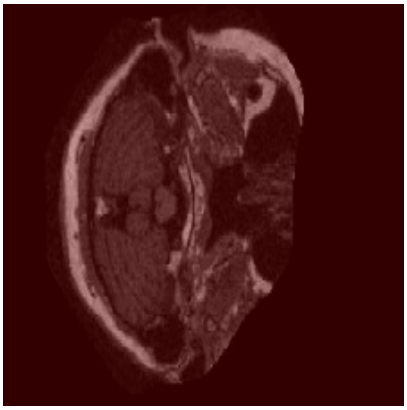
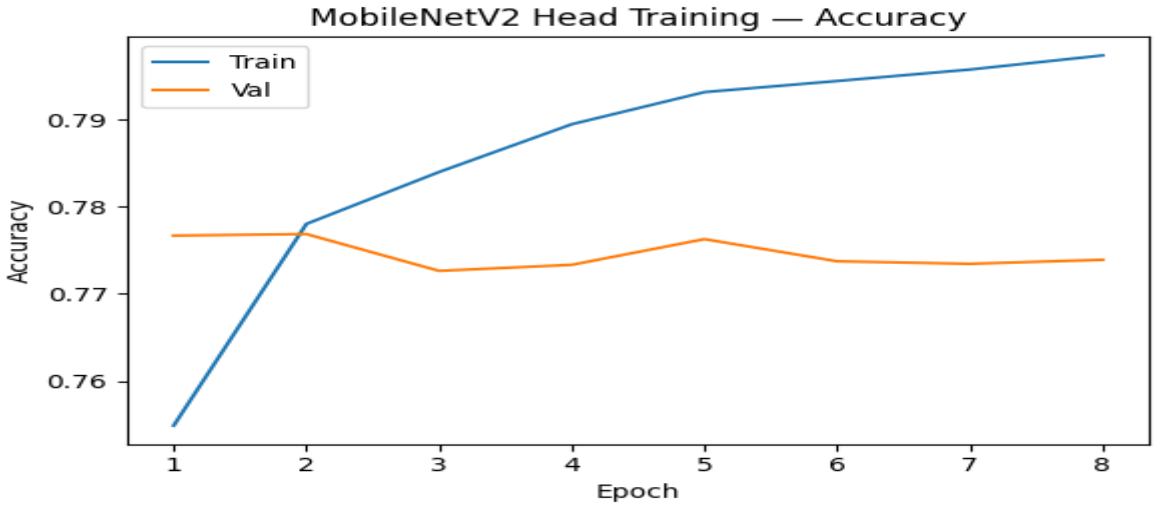
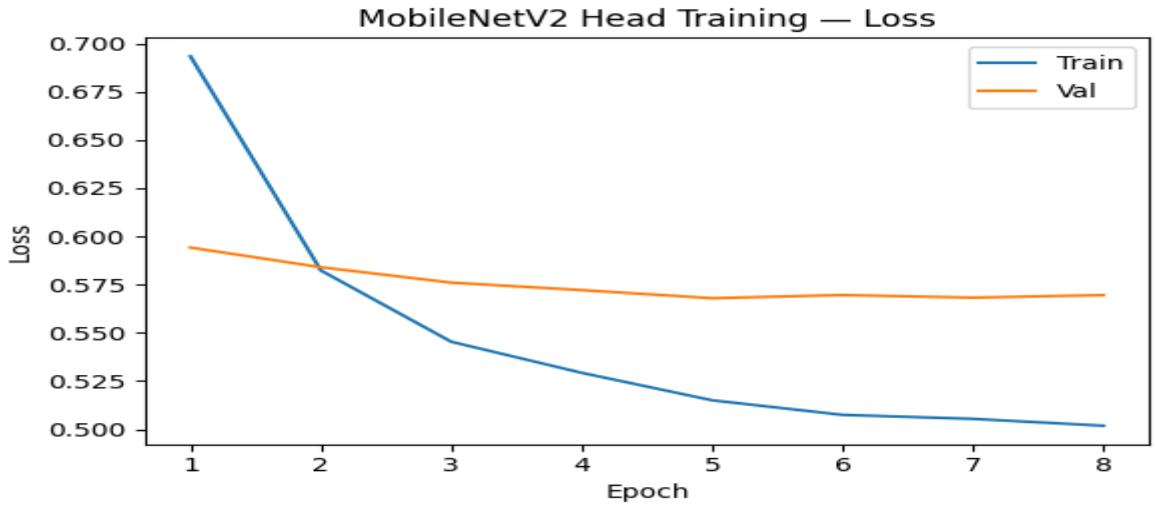

gradcam_OAS1_0003_MR1_mpr-1_104.jpg

5. Training Performance

Training and validation graphs for head training and fine-tuning (if available).

Head Training (initial)

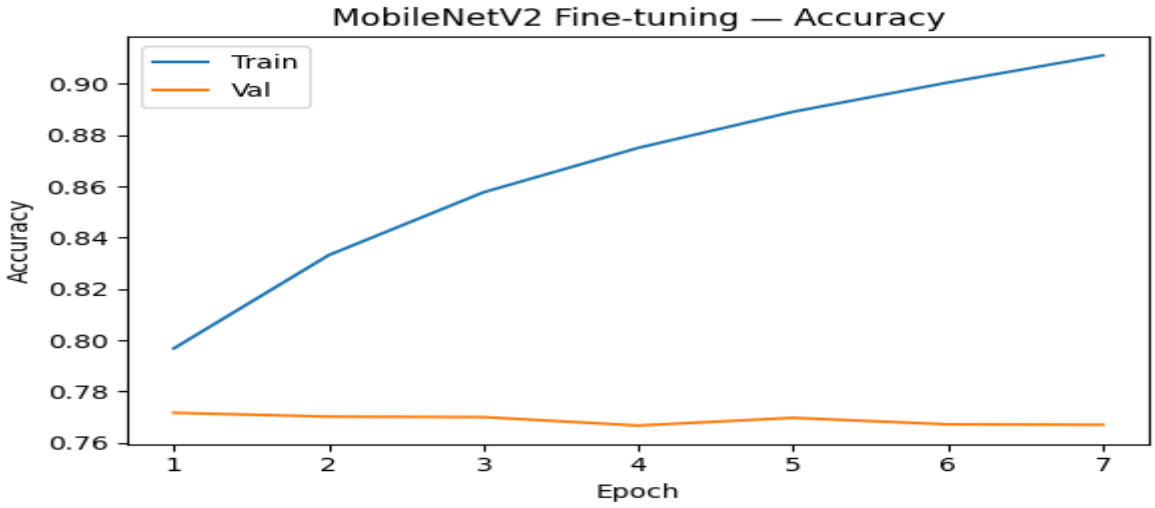


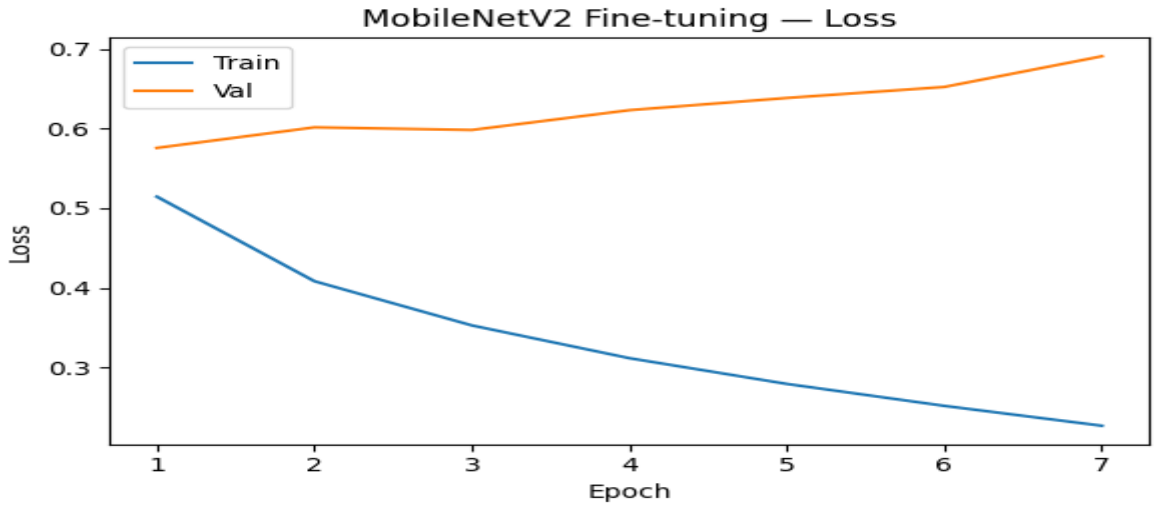


Auto-interpretation (heuristic):

- Check curves for gap between train and val accuracy (overfitting indicator).
- If validation accuracy plateaus while training accuracy rises, model may be overfitting.
- If both loss and val_loss decrease smoothly and converge, training is stable.

Fine-tuning Phase





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• Check curves for gap between train and val accuracy (overfitting indicator).
• If validation accuracy plateaus while training accuracy rises, model may be overfitting.
• If both loss and val_loss decrease smoothly and converge, training is stable.

6. Conclusions and Next Steps

- This document aggregates quantitative (classification metrics, confusion matrix) and qualitative (Grad-CAM visualizations) evidence for Phase 2 results.
- Grad-CAM maps provide interpretable activations — consider clinical validation next.
- Suggested next steps for Phase 3:
  1) Expand validation on an independent dataset and report statistical significance.
  2) Conduct ablation study (input preprocessing, augmentation, architecture choices).
  3) Deploy a small inference tool and measure inference time and resource use.
  4) Draft corresponding thesis Methods & Results sections using these visuals.